

Student ID:	24101226	Lab Section:	11A
Name:	Mehedi Hasan Roudno	Lab Group:	06

Experiment-06

Study of I-V Characteristics of MOSFET and Implementation of Logic Functions

CSE251 - Electronic Devices and Circuits Lab



Objective

1. To observe and understand the I-V characteristics of MOSFET.
2. To implement a NAND gate and a Logic Function using MOSFETs and verify the truth tables.

Equipments

1. MOSFET (IRF 540) - ($\times 3$)
2. Resistances: $4.7\text{ k}\Omega$ ($\times 1$), $100\text{ k}\Omega$ ($\times 1$)
3. DC power supply
4. Oscilloscope
5. Trainer Board
6. Breadboard
7. Chords and Wire

Background Theory

Introduction to MOSFET

Three-terminal devices are far more useful than two-terminal ones (such as Diodes) because they can be used in a multitude of applications, ranging from signal amplification to digital logic and memory. The basic principle involved is the use of the voltage between two terminals to control the current flowing in the third terminal. In this way a three-terminal device can be used to realize a controlled source.

The control signal can be used to cause the current in the third terminal to change from zero to a large value, thus allowing the device to act as a switch. Switch is the basis for the realization of the logic inverter, which is a basic element of digital circuits.

There are two major types of three-terminal semiconductor devices: (i) MOSFET (Metal-Oxide Semiconductor Field-Effect Transistor) and (ii) BJT (Bipolar Junction Transistor). Although both of them offer unique features and areas of application, MOSFET has become by far the most widely used electronic device, especially in the design of integrated circuits (ICs).

There are two kinds of MOSFET: (i) NMOS and (ii) PMOS. In this experiment, we will study about the I-V characteristics of NMOS and design an NAND gate and a Boolean Logic Function using the NMOS transistor. The figure above shows the IC and the circuit diagram of a MOSFET (NMOS).

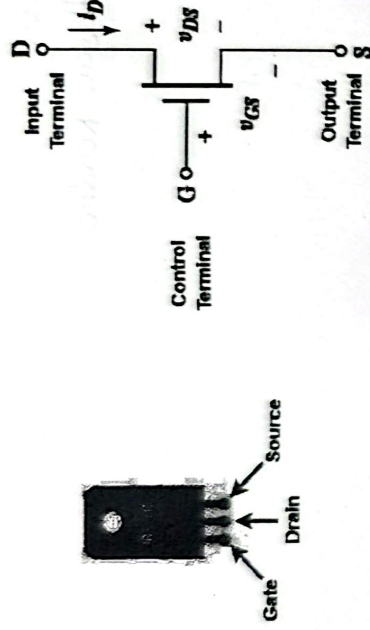


Figure 1: NMOS IC (IRF540) and Circuit Diagram

MOSFET as a Switch

MOSFET is a three-terminal device with a control terminal, an input terminal, and an output terminal. The control terminal of the MOSFET is called its gate G, the input terminal its drain D, and the output terminal its source S. MOSFET can act as a switch depending upon the applied voltage in the input terminal. A simple circuit model known as 'Switch Model' or 'S-Model' is used to model this behavior.

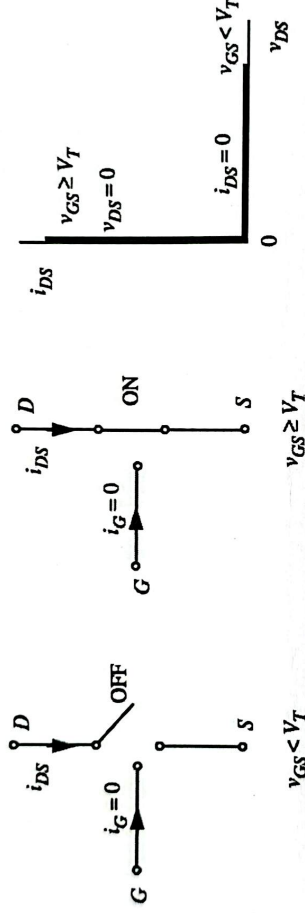


Figure 2: S-Model of MOSFET

Note that the current $I_G = 0$. We can summarize the S-model of the MOSFET in algebraic form as:

when $\rightarrow v_{GS} < V_T$, then $\rightarrow i_{DS} = 0$ and when $\rightarrow v_{GS} \geq V_T$, then $\rightarrow v_{DS} = 0$

MOSFET Switch Implementation of Logic Gates and Logic Functions

Switches can be used to build logic gates. As we can use MOSFET as a switch, logic gates can be built using MOSFETs. The following subsections discuss two logic gates, NOT gate and NAND gate, which are implemented using MOSFETs. We can also implement various Logic Functions using MOSFETs. To demonstrate this thing, a logic function will be implemented in Task-01.

NOT Gate/Inverter Using MOSFET

Consider the circuit shown in following figure, which comprises a MOSFET and a load resistor powered by a supply voltage V_S . This circuit acts as a NOT Gate/Inverter.

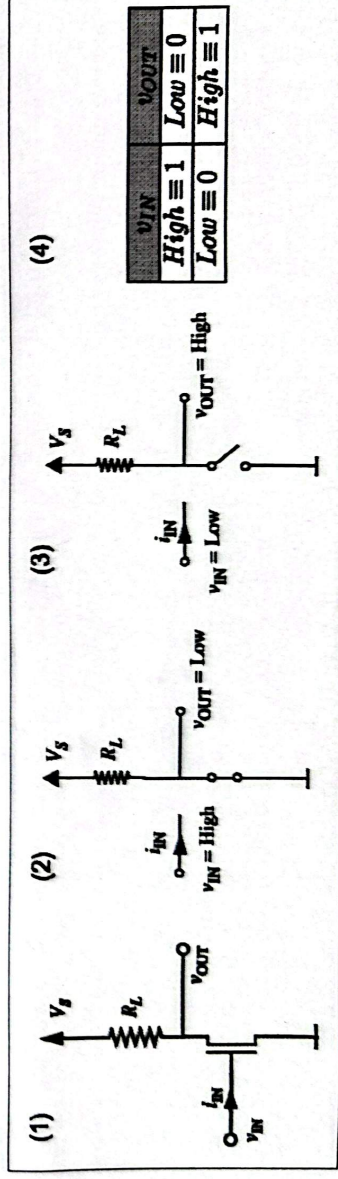


Figure 3: (1) MOSFET NOT Gate (2) When Input is '1' (3) When Input is '0' (4) Truth Table

NAND Gate Using MOSFET

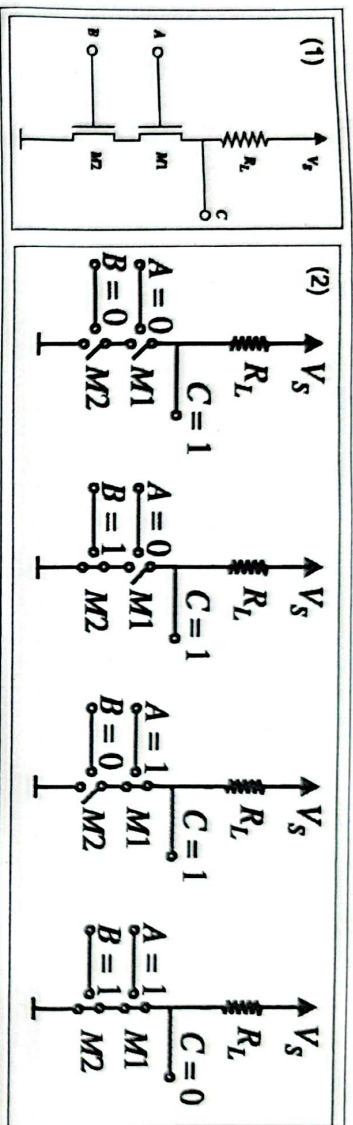


Figure 4: (1) MOSFET NAND Gate (2) Equivalent Circuits for Different Inputs

Consider the circuit shown in the figure above, which comprises a two MOSFETs and a load resistor powered by a supply voltage V_S . This circuit acts as a NAND gate. NAND gate is a universal gate that can be used to implement any Logic Function.

Logic Function Using MOSFET

We can use combinations of MOSFETs to implement various logic functions. The circuit shown in Figure-5 implements $f = \overline{xy} + z$.

I-V Characteristics of a MOSFET

The S-model of a MOSFET is a simplification of the actual I-V characteristics of a MOSFET which helps us understand the behavior of a MOSFET as a switch. But the actual I-V characteristics is very different from it.

Figure-6 represents the actual I-V characteristics of a MOSFET. There are 3 operating regions in the I-V characteristics graph: **Cutoff, Triode, Saturation**. The first two are useful if we want to use MOSFET as a switch. On the other hand, if we want to use MOSFET as an amplifier, we must operate it in the saturation region.

MOSFET's operating region depends on the applied voltages at Gate, Drain, Source terminals which can be seen in the I-V characteristics graph.

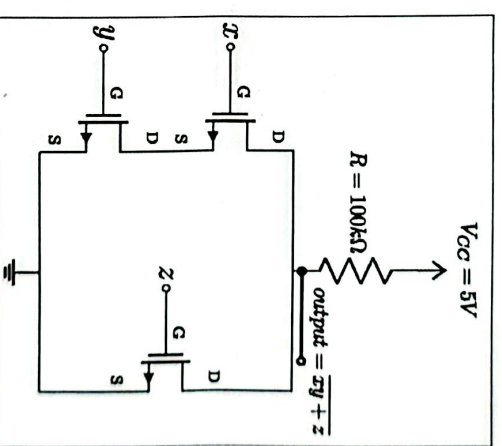


Figure 5: Logic Function using MOSFET

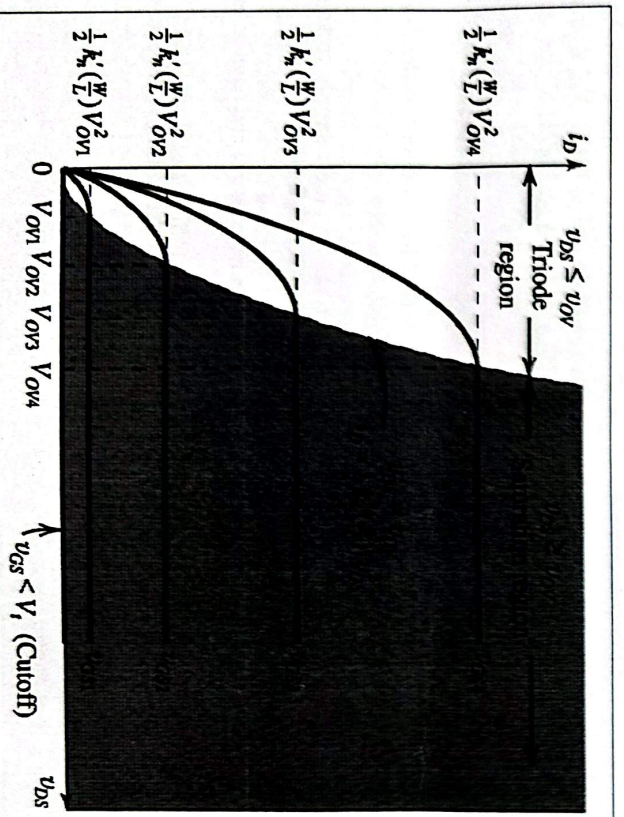


Figure 6: I-V Characteristics of a MOSFET

In this graph

- V_T = Threshold Voltage
- u_{OV} = Overdrive voltage
- L = Channel Length
- W = Channel Width
- k'_n = Process Transconductance Parameter

$$u_{GS} = u_G - u_S$$

$$u_{DS} = u_D - u_S$$

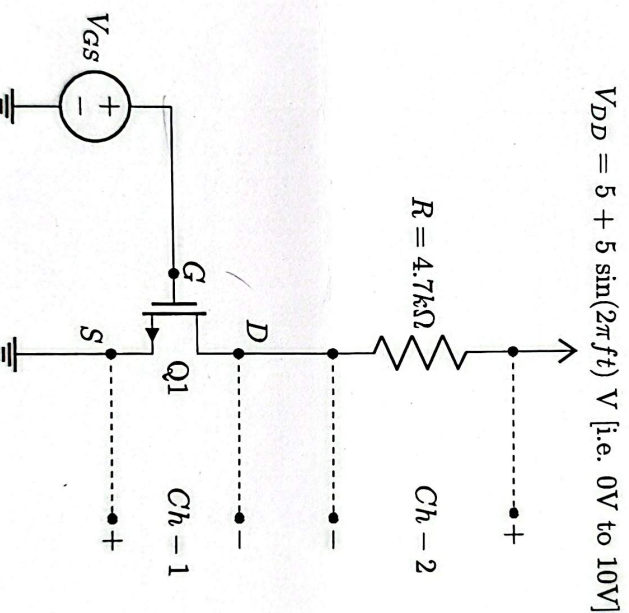
$$u_{OV} = u_{GS} - V_T$$

We can summarize the operating regions of as MOSFET as:

When	Operating Mode	Equation of Current
$v_{GS} < V_T$	Cutoff Mode	$i_D = 0$
$v_{GS} \geq V_T, v_{DS} \leq v_{OV}$	Triode Mode	$i_D = k'_n (W/L) (v_{OV} - \frac{1}{2} v_{DS}) v_{DS}$
$v_{GS} \geq V_T, v_{DS} > v_{OV}$	Saturation Mode	$i_D = \frac{1}{2} k'_n (W/L) v_{OV}^2$

The value of V_T is controlled during device fabrication and typically lies in the range of 0.3 V to 1 V. From the graph, we see that, for the same value of v_{DS} , i_{DS} increases if we increase v_{GS} . Also, for a lower value of v_{GS} , the MOSFET operates in Saturation for a lower value of v_{DS} and for higher value of v_{GS} , the MOSFET goes into Saturation for a higher value of v_{DS} .

Task-01: I-V Characteristics of a MOSFET



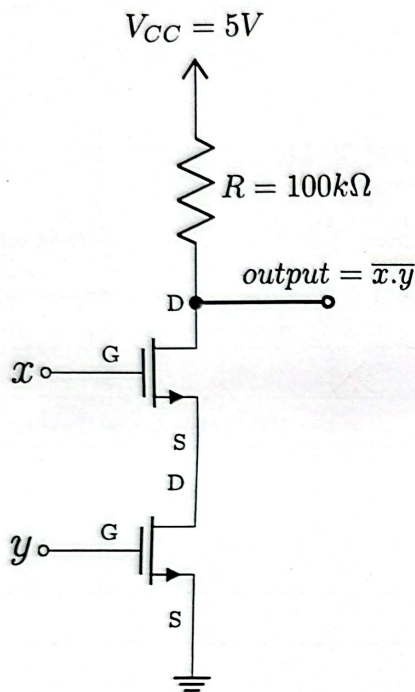
Procedure

1. Construct the circuit shown above. Use a data switch and connect it to the Gate of the MOSFET which will provide necessary voltage for V_{GS} . Use the function generator for $V_{DD} = 5 + 5 \sin(2\pi ft)$ V [i.e. an AC voltage that varies from 0V to 10V] and set $f = 50$ Hz.
2. Set the oscilloscope in X-Y mode. Invert the Channel-1.
3. Observe the plot in the oscilloscope when the data switch is ON and OFF. This plot shows the I-V characteristics of a MOSFET as a switch. Capture the plots using your mobile camera.
4. Now, disconnect the data switch from the gate of the MOSFET and connect the dc power supply to the gate terminal so that we can increase or decrease V_{GS} .
5. Rotate the voltage knob of the dc power supply slowly from 0V to 5V. You should observe the change in the I-V characteristics.
6. Use your mobile camera to capture the image of the I-V characteristics graphs for 3 different V_{GS} . Measure the values of V_{GS} of the captured images and write them in Data Table 1.

Data Table 1: Different Values of V_{GS} to observe the change in MOSFET I-V Characteristics

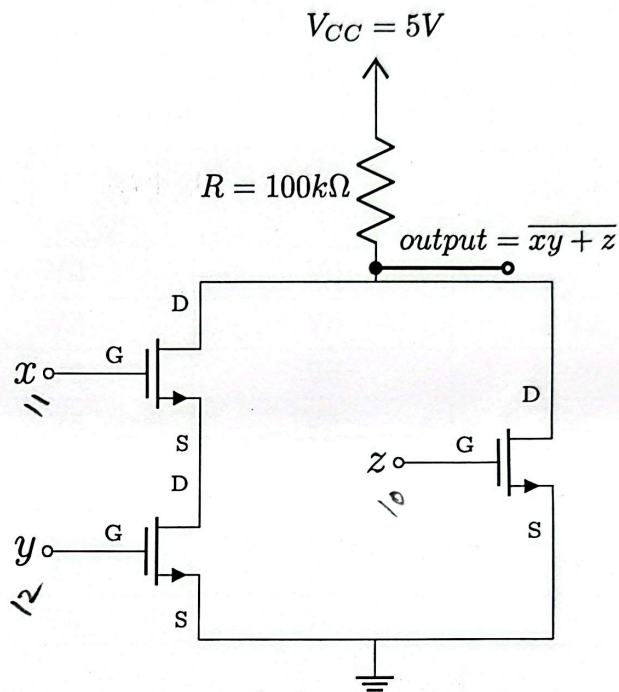
Values of V_{GS}	
1st Value	0V
2nd Value	4.5V
3rd Value	7.5V

Task-02: Logic Gate and Logic Function Implementation



Circuit-1

NAND Gate using MOSFET



Circuit-2

Logic Function, $f = \overline{xy} + z$ using MOSFET

Procedure

1. On a trainer board, setup the Circuit 1.
2. Connect the gate terminals (input x and y) to data switches. Those switches provide 5V approximately.
3. Keeping V_{CC} constant at 5V, at first turn off the data switches connected to the gate terminal. This implies you are now applying $V_x = 0V$, $V_y = 0V$. Measure the corresponding output voltage, V_{out} which should be approximately 5V which corresponds to boolean 1.
4. The boolean outputs can also be determined by the state of an LED. Connect V_{out} to one of the LEDs and check it. When the LED is ON, the boolean output is 1. Similarly, when the LED is OFF, the boolean output is 0.
5. Next, use the input voltage combinations of Data Table 1 and observe the state of LED again.
6. Verify the truth table of the NAND gate.
7. Now we will implement a Logic Function, $f = \overline{xy} + z$ using MOSFET. Circuit 2 in the figure shown above represents the Logic Function $f = \overline{xy} + z$. Setup Circuit 2 in a similar way to the procedure followed for Circuit 1.

8. Keeping V_{DD} constant at 5V, observe the state of the LED for different combinations of w and z as shown in Data Table 1.

9. Verify the truth table of the Logic Function, $f = \overline{xy} + z$.

Data Table 2: Verification of the Truth Tables of Logic Gate and Logic Function

1. NAND Gate

Input Voltage, V_x (volt)	Input Voltage, V_y (volt)	State of LED (On/Off)	Boolean Output (0 or 1)
0V	0V	On	1
0V	5V	On	1
5V	0V	On	1
5V	5V	Off	0

2. Logic Function, $f = \overline{xy} + z$

Input Voltage, V_x (volt)	Input Voltage, V_y (volt)	Input Voltage, V_z (volt)	State of LED (On/Off)	Boolean Output (0 or 1)
0V	0V	0V	On	1
0V	0V	5V	Off	0
0V	5V	0V	On	1
0V	5V	5V	Off	0
5V	0V	0V	On	1
5V	0V	5V	Off	0
5V	5V	0V	Off	0
5V	5V	5V	Off	0

Zayed Anis
24/06
Signature of the lab faculty

Report Submission Guidelines

1. Attach the signed Data Sheet (if any)
2. Attach the captured images (if any)
3. Answer the questions in the "Test Your Understanding" section
4. Add a brief Discussion regarding the experiment. For the Discussion part of the lab report, you should include the answers of the following questions in your own words:
 - What did you learn from this experiment?
 - What challenges did you face and how did you overcome the challenges? (if any)
 - What mistakes did you make and how did you correct the mistakes? (if any)
 - How will this experiment help you in future experiments of this course?

Test Your Understanding

Answer the following questions:

1. We can use the MOSFET as a switch. Which operating regions do we need for this purpose and why?

Answer:

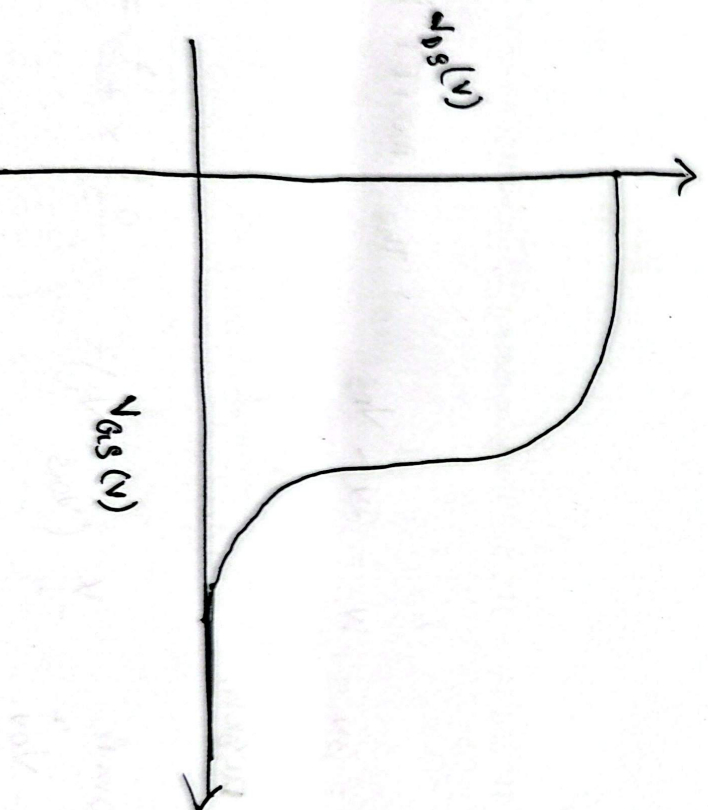
Yes, we can use MOSFET as a switch in two regions,

- ① Cut off region: When the V_{GS} is less than V_T , the MOSFET is off. It acts like an open switch with very high resistance between drain and source.

- ② Triode region: $V_{GS} \geq V_T$, the MOSFET is ON. It acts like a closed switch with a very low resistance between drain and source, allowing current to flow.

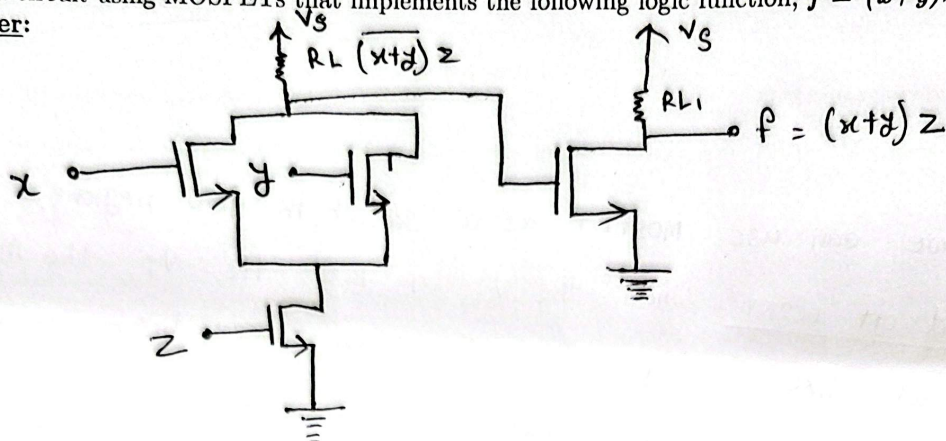
2. Draw the VTC (V_{DS} vs V_{GS}) graph of a MOSFET operating as a switch.

Answer:



3. Draw a circuit using MOSFETs that implements the following logic function, $f = (x+y)z$

Answer:



4. When a MOSFET is in triode mode, does it display zero, finite or infinite resistance? Justify your answer.

Answer:

When a MOSFET is in triode mode, it displays a finite resistance.

$$V_{DS} \leq V_{GS} - V_T ; i_D = k_n \left(\frac{W}{L} \right) \left(V_{OV} - \frac{1}{2} V_{DS} \right) V_{DS}$$

for a small value of V_{DS} , the quadratic eq of i_D become negligible. Then we can use ohms law,

$$R = \frac{V_{DS}}{I_D}$$

5. If $V_{DS} = 3V$, $V_{GS} = 3V$ and $V_T = 1V$, find the gate current, drain current and source current of a MOSFET given that $k = 0.5mA/V^2$.

Answer:

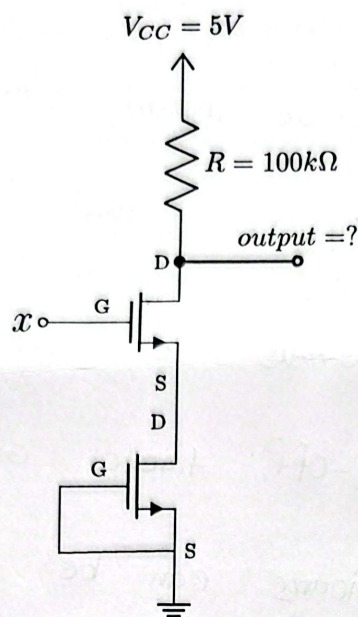
$V_{GS} - V_T = 3V - 1V = 2V < V_{DS}$. The MOSFET is in saturation region.

$$I_G = 0mA$$

$$I_D = \frac{k}{2} V_{OV}^2 = \frac{k}{2} (V_{GS} - V_T)^2 = \frac{0.5}{2} \times 2^2 = 1mA$$

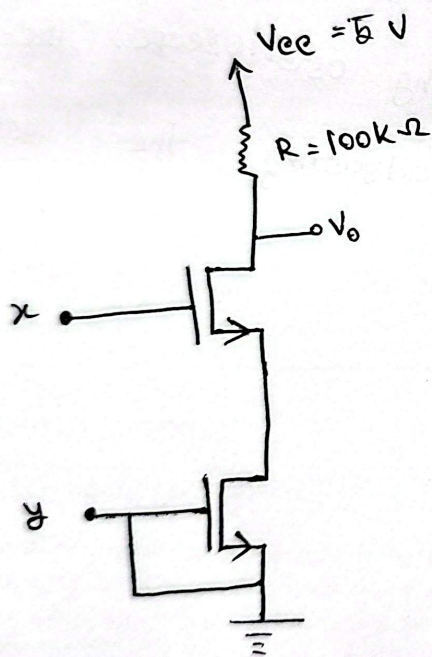
$$I_S = I_D = 1mA$$

$$I_G = 0mA, I_S = I_D = 1mA$$



6. What is the value of the output of the circuit above?

Answer:



As y is connected to ground, $y = 0V$

x	y	V_o
0	0	0 (0V)
1	0	1 (5V)

Discussion :

In this experiment, we learnt how to analyze the I-V characteristics of MOSFET and implement basic logic functions. It helped us to understand how a MOSFET operates in cut-off, triode and saturation regions and how these behaviours can be used to design switching and logic design circuits.

One issue we faced was interpreting IV curves due to difficulty of using oscilloscope. The signals were not clear. After adjusting the settings, we were able to interpret.

GWINSTEK

10k pts

200kSa/s



Trig'd

1

2

1



F 50.0048Hz

1 ↓ = 5.00V 2 = 5.00V

5ms 0.00000s

1 ↑ 5V DC

Coupling
DC AC GND

Impedance
1MΩ

Invert
On Off

Bandwidth
Full

Expand
By Ground

Position /
Set to 0
0.000V

Probe
Voltage
1 X

GWINSTEK

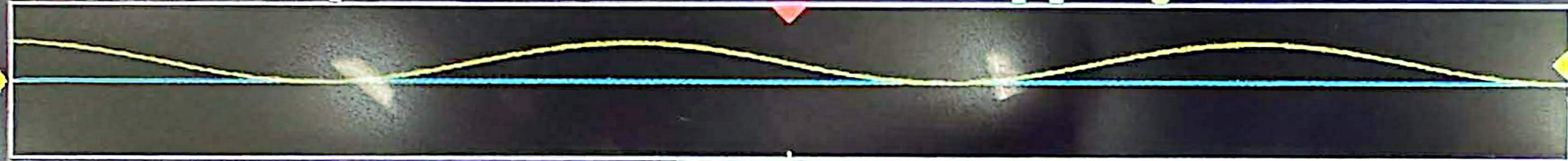
10k pts

200kSa/s



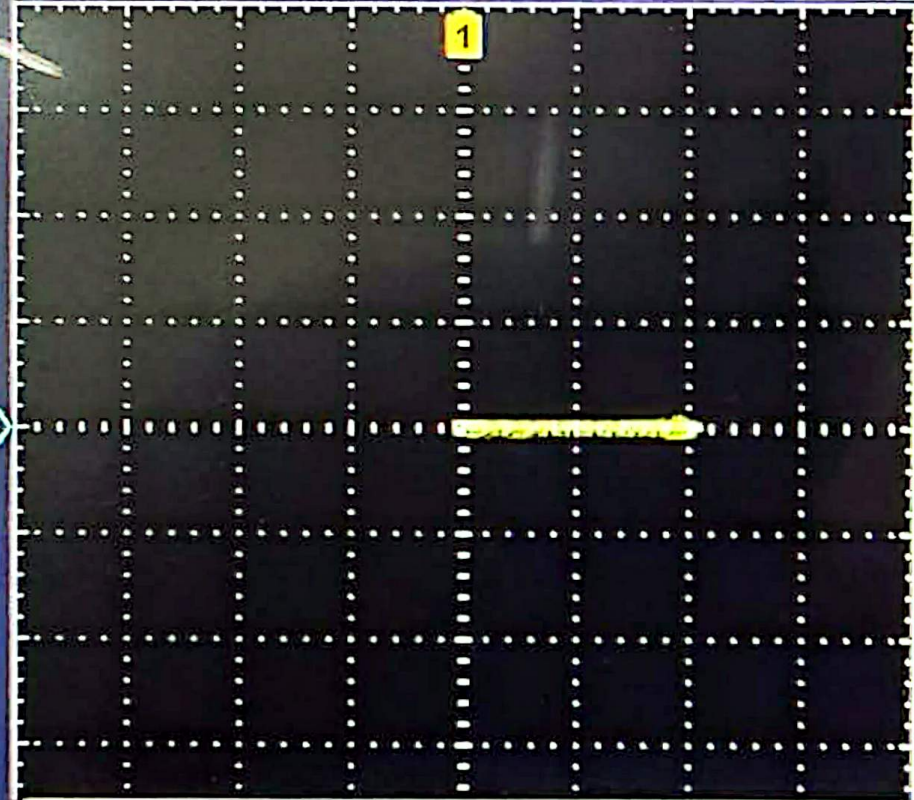
Trig'd

1



1

2



F 50.0104Hz

1 ↓ = 5.00V

2 = 5.00V

5ns

0.00000s

1

5V

DC

Coupling
DC AC GND

Impedance
1MΩ

Invert
On Off

Bandwidth
Full

Expand
By Ground

Position /
Set to 0
0.000V

Probe
Voltage
1 X

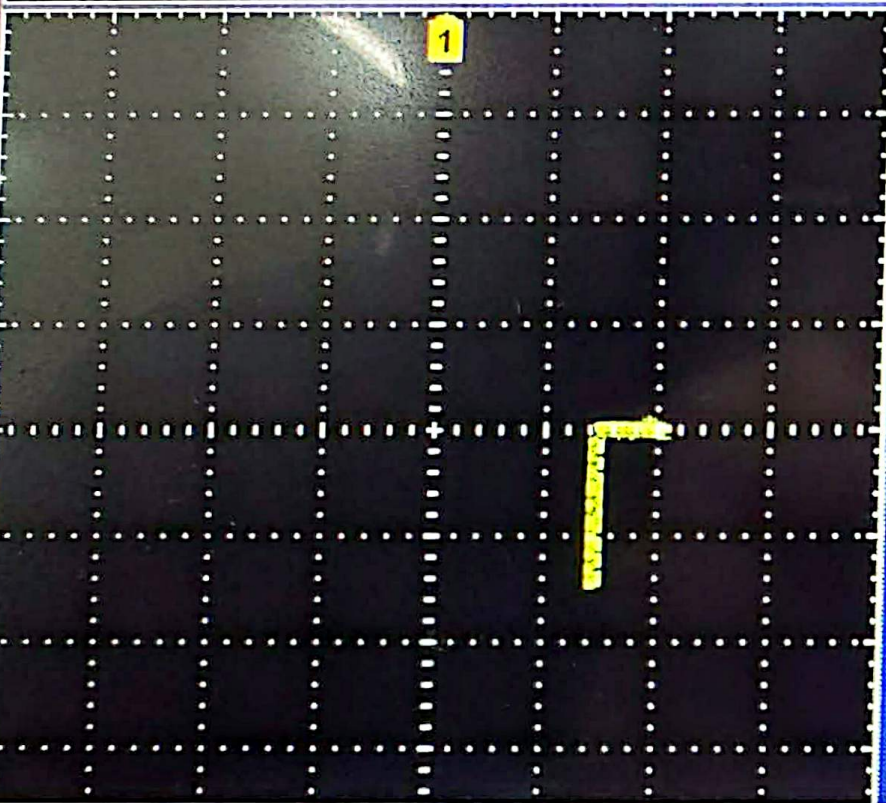
WINSTEK

10k pts

200kSa/s



Auto



1 ↓ = 5.00V 2 = 5.00V

5ms

0.00000s

F <2Hz

1 ↑

5V

DC

Coupling
DC AC GND

Impedance
1MΩ

Invert
On Off

Bandwidth
Full

Expand
By Ground

Position /
↓ Set to 0
0.000V

Probe
Voltage
1 X